

Тема: Настройка протокола STP (IEEE 802.1D)

Все команды для настройки включаются в отчет в текстовом виде, не скриншоты.

nb! - отметка в тексте, "обратите особое внимание"

Полезная информация: схема сохранена на сервере в проекте под именем Menzhulin-lab2-template, можно использовать кнопку Duplicate

1) Для заданной на схеме schema-lab2 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета

Первый:

```
vIOS-L2-01>enable
```

```
vIOS-L2-01#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
vIOS-L2-01(config)#hostname Layer2Switch-1
```

```
Layer2Switch-1(config)#spanning-tree mode pvst
```

```
Layer2Switch-1(config)#spanning-tree vlan 1 priority 4096
```

```
Layer2Switch-1(config)#end
```

```
Layer2Switch-1#write memory
```

Building configuration...

Compressed configuration from 5281 bytes to 2027 bytes[OK]

Остальные:

```
vIOS-L2-01>enable
```

```
vIOS-L2-01#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
vIOS-L2-01(config)#hostname Layer2Switch-2
```

```
Layer2Switch-2(config)#spanning-tree mode pvst
```

```
Layer2Switch-2(config)#end
```

```
Layer2Switch-2#write memory
```

Building configuration...

Compressed configuration from 5244 bytes to 2004 bytes[OK]

2) Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать

PC1> ip 192.168.1.1 255.255.255.0

Checking for duplicate address...

PC1 : 192.168.1.1 255.255.255.0

PC2> ip 192.168.1.2 255.255.255.0

Checking for duplicate address...

PC2 : 192.168.1.2 255.255.255.0

PC3> ip 192.168.1.3 255.255.255.0

Checking for duplicate address...

PC3 : 192.168.1.3 255.255.255.0

PC4> ip 192.168.1.4 255.255.255.0

Checking for duplicate address...

PC4 : 192.168.1.4 255.255.255.0

PC5> ip 192.168.1.5 255.255.255.0

Checking for duplicate address...

PC5 : 192.168.1.5 255.255.255.0

PC6> ip 192.168.1.6 255.255.255.0

Checking for duplicate address...

PC6 : 192.168.1.6 255.255.255.0

PC1:

PC1> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=3.796 ms

84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=4.966 ms

84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=1.070 ms

84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=1.059 ms

84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=4.356 ms

PC1> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=9.798 ms

84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=3.039 ms

84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=6.267 ms

84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=2.565 ms

84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=5.299 ms

PC1> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=11.155 ms

84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=2.993 ms

84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=6.254 ms

84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=4.772 ms

84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=6.647 ms

PC1> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=8.485 ms

84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=6.582 ms

84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=6.491 ms

84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=2.000 ms

84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=6.251 ms

PC1> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=15.625 ms

84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=7.672 ms

84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=2.817 ms

84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=9.382 ms

84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=3.441 ms

PC 2:

PC2> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=10.470 ms

84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=8.991 ms

84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=6.042 ms

84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=8.062 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=10.233 ms

PC2> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=9.676 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=8.353 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=6.126 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=9.040 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=5.769 ms

PC2> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=11.384 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=1.886 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=4.042 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=2.233 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=2.299 ms

PC2> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=7.200 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=8.837 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=10.261 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=3.946 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=7.699 ms

PC 3:

PC3> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=5.561 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=3.490 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=1.998 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=6.776 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=7.899 ms

PC3> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=12.597 ms

84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=4.510 ms

84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=7.551 ms

84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=7.359 ms

84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=4.391 ms

PC3> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=14.861 ms

84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=6.217 ms

84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=9.172 ms

84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=6.135 ms

84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=16.554 ms

PC 4:

PC4> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=2.493 ms

84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=2.729 ms

84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=8.770 ms

84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=5.047 ms

84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=6.681 ms

PC4> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=7.532 ms

84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=4.976 ms

84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=6.183 ms

84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=14.486 ms

84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=7.387 ms

PC 5:

PC5> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=0.968 ms

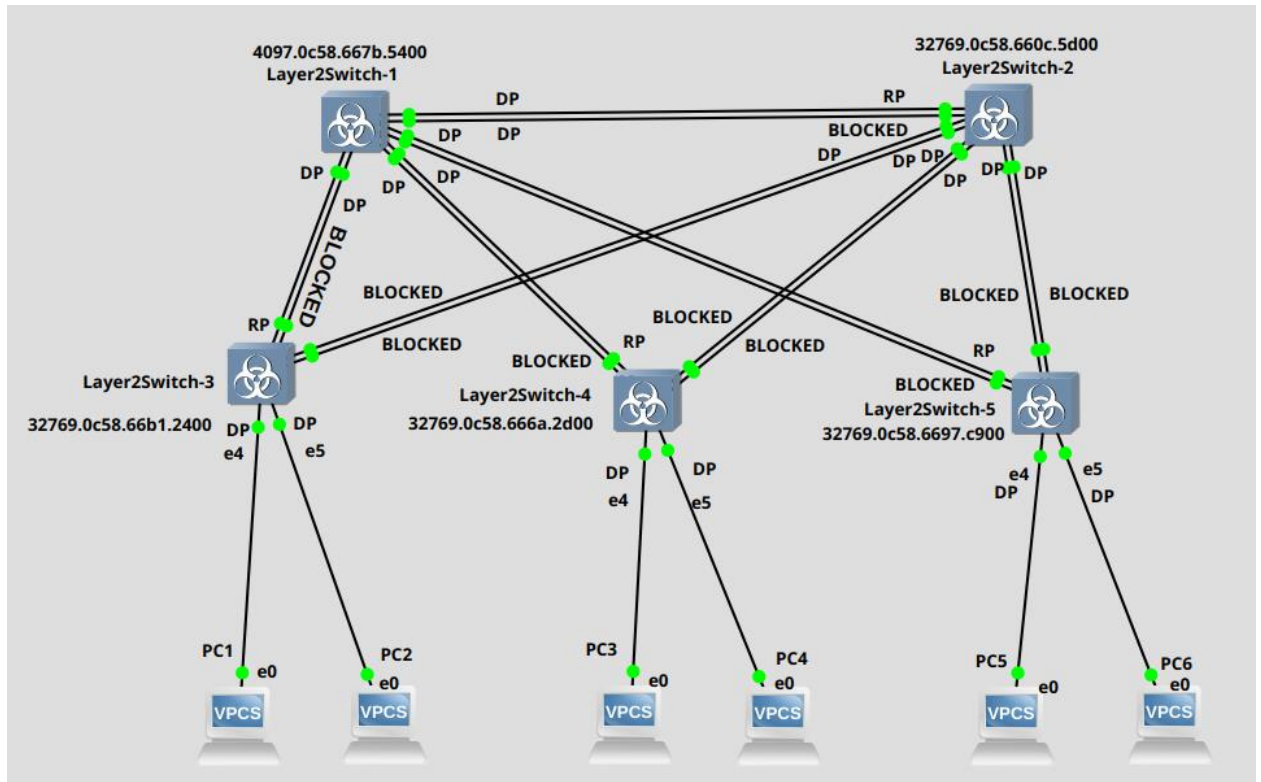
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=5.136 ms

84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=3.373 ms

84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=3.850 ms

84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=1.207 ms

3) На изображении схемы отметить BID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл



4) При помощи wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет

stp										
Time	Source	Destination	Protocol	Length	Info					
1 0.000000	0c:58:66:7b:54:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:58:66:7b:54:00 Cost = 0 Port = 0x0002					
2 3.004710	0c:58:66:7b:54:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:58:66:7b:54:00 Cost = 0 Port = 0x0002					
3 1.392289	0c:58:66:7b:54:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:58:66:7b:54:00 Cost = 0 Port = 0x8002					
4 1.205323	0c:58:66:7b:54:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:58:66:7b:54:00 Cost = 0 Port = 0x8002					
5 1.208309	0c:58:66:7b:54:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:58:66:7b:54:00 Cost = 0 Port = 0x8002					
7 1.918139	0c:58:66:0c:5d:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:58:66:0c:5d:00 Cost = 0 Port = 0x8002					
8 1.920100	0c:58:66:0c:5d:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:58:66:0c:5d:00 Cost = 0 Port = 0x8002					
9 1.925064	0c:58:66:0c:5d:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:58:66:0c:5d:00 Cost = 0 Port = 0x8002					
Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0					0000	01 00 c2 00 00 00 00 00 66 7b 54 01 00 00 42 42	X r t t	00 B		
IEEE 802.3 Ethernet					0010	03 00 00 00 00 00 01 0c 58 66 7b 54 00 00 00	X r t	00 B		
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)					0020	00 00 01 01 0c 58 66 7b 54 00 80 02 00 00 14 00	X f	00 B		
.....0. = LG bit: Globally unique address (factory default)					0030	02 00 0f 00 00 00 00 00 00 00 00 00		00 B		
.....1. = IG bit: Group address (multicast/broadcast)										
Source: 0c:58:66:7b:54:01 (0c:58:66:7b:54:01)										
.....0. = LG bit: Globally unique address (factory default)										
.....0. = IG bit: Individual address (unicast)										
Length: 38										
Padding: 0000000000000000										
[Stream index: 0]										
Logical-Link Control										
DSAP: Spanning Tree BPDU (0x42)										
0100 001. = SAP: Spanning Tree BPDU										
.....0 = IG Bit: Individual										
SSAP: Spanning Tree BPDU (0x42)										
0100 001. = SAP: Spanning Tree BPDU										
.....0 = CR Bit: Command										
Control field: 0, FuncUII (0x03)										
000. 00.. = Command: Unnumbered Information (0x00)										
....11 = Frame type: Unnumbered frame (0x3)										
Spanning Tree Protocol										
Protocol Identifier: Spanning Tree Protocol (0x0000)										
Protocol Version Identifier: Spanning Tree (0)										
BPDU Type: Configuration (0x00)										
BPDU flags: 0x00										
0... = Topology Change Acknowledgment: No										
....0 = Topology Change: No										
Root Identifier: 4096 / 1 / 0c:58:66:7b:54:00										
Root Path Cost: 0										
Bridge Identifier: 4096 / 1 / 0c:58:66:7b:54:00										
Port identifier: 0x8002										
Message Age: 0										
Max Age: 20										
Hello Time: 2										
Forward Delay: 15										

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stp						
Time	Source	Destination	Protocol	Length	Info	
3.009641	0c:58:66:7b:54:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:58:66:7b:54:00 Cost = 0 Port = 0x8003	
4.009480	0c:58:66:7b:54:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:58:66:7b:54:00 Cost = 0 Port = 0x8003	
4.382648	0c:58:66:b1:24:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:58:66:b1:24:00 Cost = 0 Port = 0x8001	
4.383497	0c:58:66:b1:24:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:58:66:b1:24:00 Cost = 0 Port = 0x8001	
4.384457	0c:58:66:b1:24:00	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:58:66:b1:24:00 Cost = 0 Port = 0x8001	
4.983931	0c:58:66:7b:54:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:58:66:7b:54:00 Cost = 0 Port = 0x8003	
4.990887	0c:58:66:7b:54:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:58:66:7b:54:00 Cost = 0 Port = 0x8003	
4.999305	0c:58:66:7b:54:02	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:58:66:7b:54:00 Cost = 0 Port = 0x8003	
Frame 20: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0				0000 01 80 c2 00 00 00 0c 58 66 7b 54 02 80 20 42 42X r(T,00B		
> IEEE 802.3 Ethernet				0010 03 00 00 00 00 00 10 01 0c 58 66 7b 54 00 00 00Xf(T...		
> Logical-Link Control				0020 00 00 10 01 0c 58 66 7b 54 00 80 03 00 00 14 00Xf(T.....		
Spanning Tree Protocol				0030 02 00 0f 00 00 00 00 00 00 00 00 00 00 00 00		
Protocol Identifier: Spanning Tree Protocol (0x0000)						
Protocol Version Identifier: Spanning Tree (0)						
BPDU Type: Configuration (0x00)						
BPDU Flags: 0x00						
> Root Identifier: 4096 / 1 / 0c:58:66:7b:54:00						
Root Path Cost: 0						
> Bridge Identifier: 4096 / 1 / 0c:58:66:7b:54:00						
Port Identifier: 0x8003						
Message Age: 0						
Max Age: 20						
Hello Time: 2						
Forward Delay: 15						

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Time	Source	Destination	Protocol	Length	Info
3.001078	0c:58:66:7b:54:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:58:66:7b:54:00 Cost = 0 Port = 0x8006
4.999931	0c:58:66:7b:54:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:58:66:7b:54:00 Cost = 0 Port = 0x8006
4.520836	0c:58:66:7b:54:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:58:66:7b:54:00 Cost = 0 Port = 0x8006
4.533838	0c:58:66:7b:54:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:58:66:7b:54:00 Cost = 0 Port = 0x8006
4.536942	0c:58:66:7b:54:05	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:58:66:7b:54:00 Cost = 0 Port = 0x8006
4.555839	0c:58:66:6a:2d:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:58:66:6a:2d:00 Cost = 0 Port = 0x8002
4.558194	0c:58:66:6a:2d:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:58:66:6a:2d:00 Cost = 0 Port = 0x8002
4.559815	0c:58:66:6a:2d:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:58:66:6a:2d:00 Cost = 0 Port = 0x8002

> Frame 16: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0	0000 01 80 c2 00 00 00 0c 58 66 7b 54 05 00 20 42 42X f(T 0B
> IEEE 802.3 Ethernet	0010 03 00 00 00 00 00 10 01 0c 58 66 7b 54 00 00 00Xf(T...
> Logical-Link Control	0020 00 00 10 01 0c 58 66 7b 54 00 00 00 00 14 00Xf{ T.....
> Spanning Tree Protocol	0030 02 00 0f 00 00 00 00 00 00 00 00 00

```

- Protocol Identifier: Spanning Tree Protocol (0x8000)
- Protocol Version Identifier: Spanning Tree (0)
- BPDU Type: Configuration (0x00)
- BPDU Flags: 0x00
- Root Identifier: 4096 / 1 / 0c:58:66:7b:54:00
- Root Path Cost: 0
- Bridge Identifier: 4096 / 1 / 0c:58:66:7b:54:00
- Port Identifier: 0x8006
- Message Age: 0
- Max Age: 20
- Hello Time: 2
- Forward Delay: 15

```

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Time	Source	Destination	Protocol	Length	Info
168.082963	0c:58:66:7b:54:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:58:66:7b:54:00 Cost = 0 Port = 0x8008
169.085829	0c:58:66:7b:54:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:58:66:7b:54:00 Cost = 0 Port = 0x8008
169.199137	0c:58:66:7b:54:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:58:66:7b:54:00 Cost = 0 Port = 0x8008
169.215045	0c:58:66:7b:54:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:58:66:7b:54:00 Cost = 0 Port = 0x8008
169.219091	0c:58:66:7b:54:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:58:66:7b:54:00 Cost = 0 Port = 0x8008
170.061949	0c:58:66:97:c9:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:58:66:97:c9:00 Cost = 0 Port = 0x8002
170.064551	0c:58:66:97:c9:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:58:66:97:c9:00 Cost = 0 Port = 0x8002
170.064772	0c:58:66:97:c9:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:58:66:97:c9:00 Cost = 0 Port = 0x8002

> Frame 726: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0	0000 01 80 c2 00 00 00 0c 58 66 7b 54 07 00 20 42 42X f(T 0B
> IEEE 802.3 Ethernet	0010 03 00 00 00 00 00 10 01 0c 58 66 7b 54 00 00 00Xf(T...
> Logical-Link Control	0020 00 00 10 01 0c 58 66 7b 54 00 00 00 00 14 00Xf{ T.....
> Spanning Tree Protocol	0030 02 00 0f 00 00 00 00 00 00 00 00 00

```

- Protocol Identifier: Spanning Tree Protocol (0x8000)
- Protocol Version Identifier: Spanning Tree (0)
- BPDU Type: Configuration (0x00)
- BPDU Flags: 0x00
- Root Identifier: 4096 / 1 / 0c:58:66:7b:54:00
- Root Path Cost: 0
- Bridge Identifier: 4096 / 1 / 0c:58:66:7b:54:00
- Port Identifier: 0x8008
- Message Age: 0
- Max Age: 20
- Hello Time: 2
- Forward Delay: 15

```

5) Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл

Layer2Switch-3#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Layer2Switch-3(config)#interface GigabitEthernet0/0

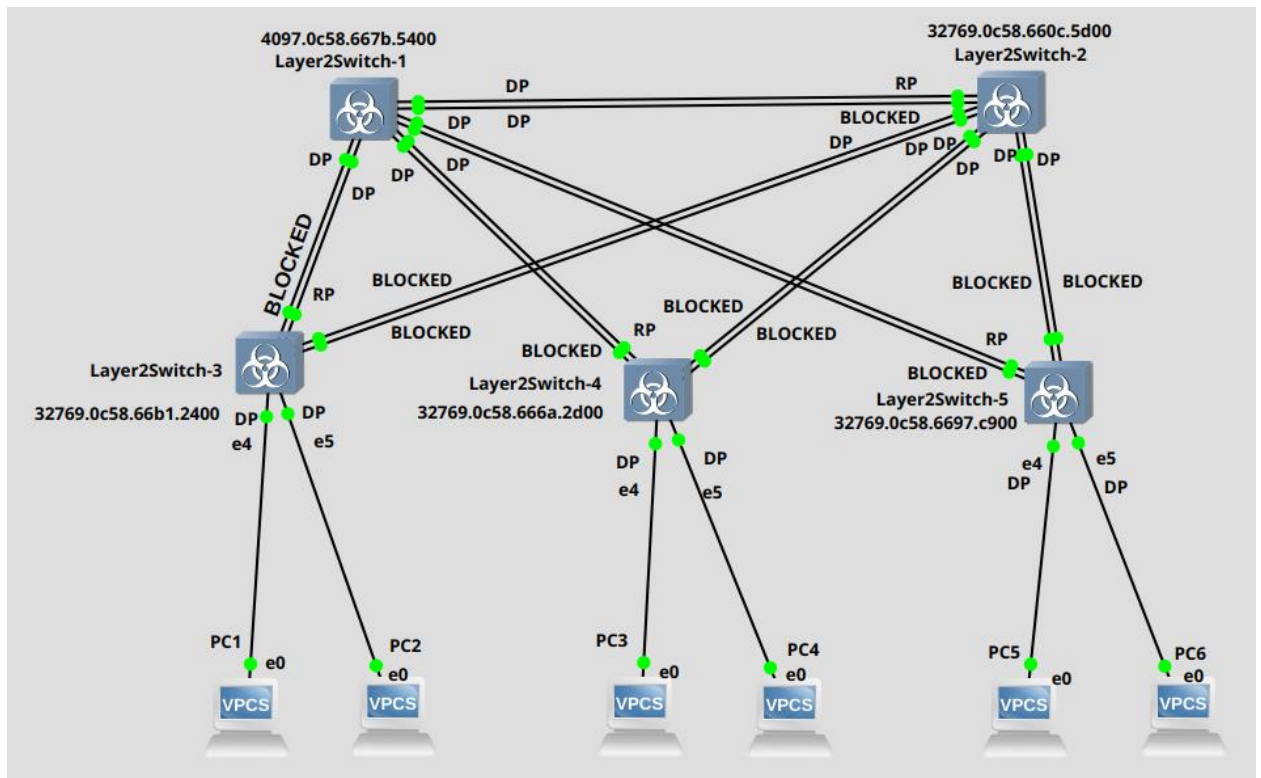
Layer2Switch-3(config-if)#spanning-tree cost 100

Layer2Switch-3(config-if)#end

Layer2Switch-3#write memory

Building configuration...

Compressed configuration from 5070 bytes to 1941 bytes[OK]



6) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств